

Introduction:

The World Happiness Report data set uses information from the Gallup World Poll to provide statistics for 149 countries in 2021. For each country, the region, happiness score, standard error, upper and lower whiskers, GDP per capita, social support score, average healthy life expectancy, freedom to make life choices score, generosity score, perception of corruption score, and various residual statistics were recorded.

The data set was analyzed to determine which regions of the world have the highest happiness score, the relationship between happiness and GDP per capita, the relationship between healthy life expectancy and happiness, and which areas of the world have the highest healthy life expectancy.

Method:

Once I imported this dataset, I took some time to review the observation ensuring that none were left blank. It was fairly clean initially, however, a large number of the columns were unrelated to my research questions so I subsetting the country name, region, happiness score, GDP per capita, and happiness life expectancy columns into a new data frame. Then, I adjusted the region column to be a factor and renamed the “United States” observation to “USA” in order to better work with the world map in R. Additionally, I merged the happiness dataset with the world map and arranged them by group and order. I was then able to create each of my plots: a map of the happiness scores across the world (visual 1), happiness score vs GDP per capita (visual 2), healthy life expectancy vs happiness score (visual 3), and a map of the healthy life expectancy across the world (visual 4).

Visual 1

```
Map of happiness scores
```{r}
ggplot (happiness.map , aes(long, lat, group = group, fill = Ladder.score)) +
 geom_polygon(color = " black ") +
 scale_fill_gradient(low = "white", high = "darkgreen")
```
```

Visual 2

```
Chart of GDP per Capita vs Happiness score
```{r}
ggplot (happiness, aes(x = Logged.GDP.per.capita, y = Ladder.score)) +
 geom_point() +
 geom_smooth(method = "lm", se=F) +
 xlab("GDP per Capita") +
 ylab("Happiness Score")
```
```

Visual 3

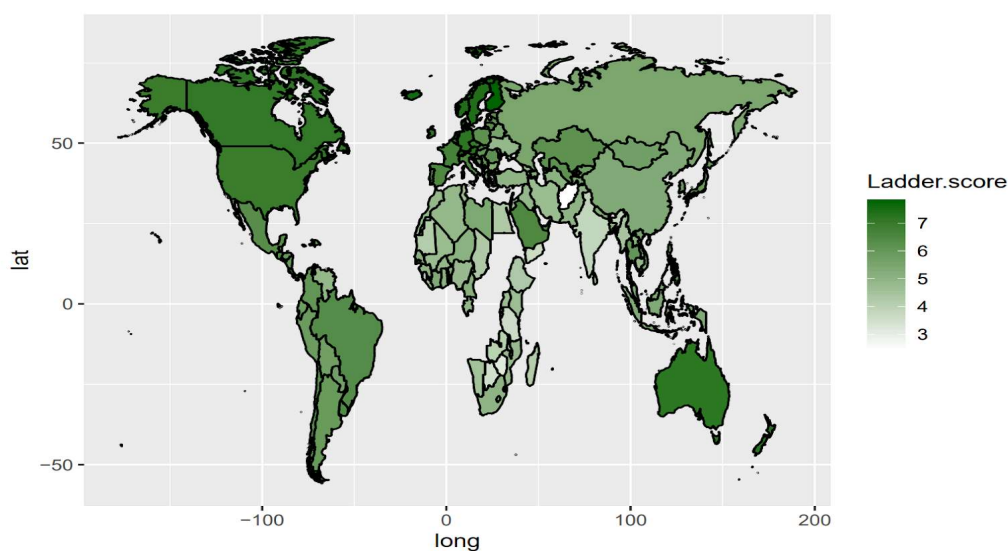
```
Chart of happiness level vs healthy life expectancy
{r}
ggplot (happiness, aes(x = Ladder.score, y = Healthy.life.expectancy)) +
  geom_point(aes(color = Regional.indicator)) +
  ylab("Healthy Life Expectancy") +
  xlab("Happiness Score")
}
```

Visual 4

```
Map of healthy life expectancy
{r}
ggplot (happiness.map , aes(long, lat, group = group, fill = Healthy.life.expectancy)) +
  geom_polygon(color = "black ") +
  scale_fill_gradient(low = "white", high = "orange" )
}
```

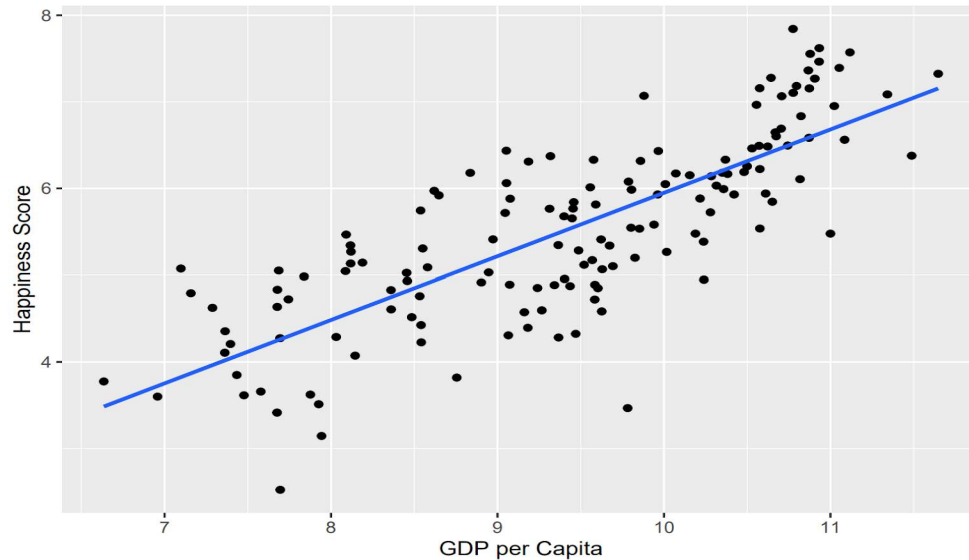
Results and Discussion:

Map of Happiness Scores



This map utilizes different levels of saturation in order to convey the reported happiness of different countries across the world. The Americas, Western Europe, and Australia appear to have the highest levels of happiness, while Africa and India appear to have the lowest.

Happiness vs GDP per capita



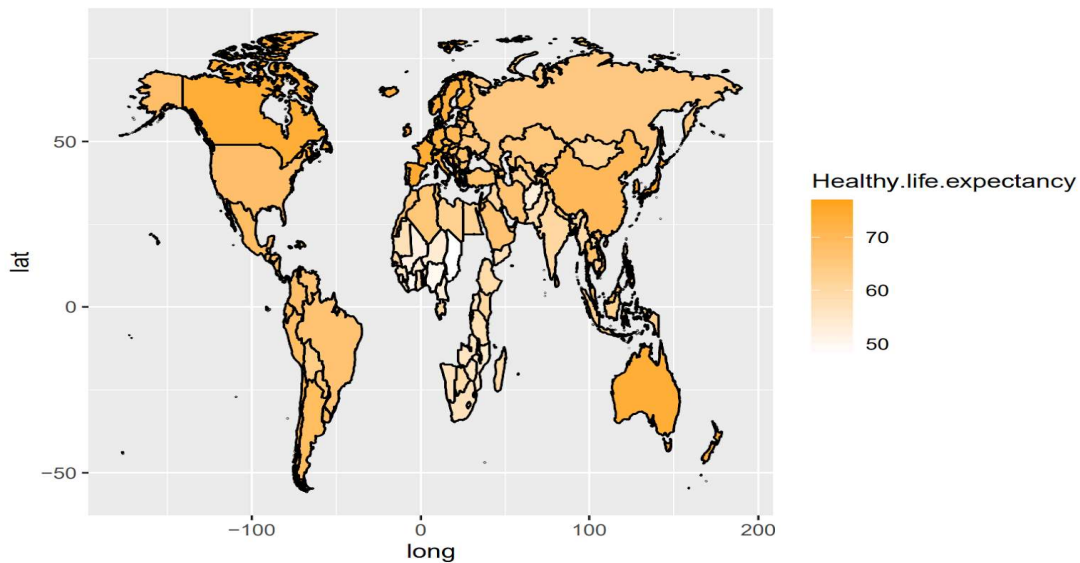
This bivariate scatterplot conveys a strong positive correlation between GDP per capita and reported happiness score. Although there is some variation, a linear line of best fit still represents the data well.

Healthy Life Expectancy vs Happiness Score



This scatter plot shows both the relationship between reported happiness and healthy life expectancy and the regional distribution of those scores. There is a strong positive correlation between those two variables and from the graph, it can be concluded that the Sub-Saharan Africa region has the lowest happiness scores and lowest life expectancy. Furthermore, Western Europe and parts of East Asia appear to have the highest happiness scores and highest life expectancy.

Map of Healthy Life Expectancy



Finally, this map utilizes different levels of saturation to convey the average healthy life expectancy in each country. At first, it appears similar to the previous map of happiness levels which makes sense as the two variables are highly correlated. However, in this case, Canada, Australia, and Western Europe appear to have the highest life expectancy while Eastern Europe and Africa have the lowest.

Conclusion:

Through the analysis of this dataset, several conclusions can be inferred. When considering the regions of the world with the highest and lowest reported happiness, we can refer to the first visual: a Map of Happiness Scores. This illustrates that the highest happiness scores reported are in North and South America, Western Europe, and Australia while Africa, India, and Eastern Europe appear to have the lowest reported happiness scores.

The second visual, Happiness vs GDP per capita, shows that there appears to be a strong positive correlation between the reported happiness score and the GDP per capita. This makes sense as higher economic security is often associated with happier citizens overall. A strong positive correlation also exists between healthy life expectancy and reported happiness scores. Thus, the countries with the lowest happiness score, such as those in Sub-Saharan Africa, also tend to have the lowest life expectancies. Additionally, the countries with the highest happiness scores, such as those in Western Europe, have the highest average life expectancies.

Finally, although the Map of Healthy Life Expectancy appears similar to the previous map of reported happiness, there are slight differences due to the imperfect correlation between the two variables. The regions with the highest healthy life expectancy are Canada, Australia, and Western Europe. Furthermore, most of Africa, Russia, and Southeast Asia have the lowest life expectancies.

Therefore we can conclude that there are strong correlations between reported happiness and both GDP per capita and healthy life expectancy. This is a logical conclusion considering economic security often leads to prolonged life expectancy and greater happiness overall.

Ideally, this data, along with that found in the original dataset, would be utilized by policymakers to make decisions with the intent of improving the happiness and well-being of their population.